IVAN MAKARENKO

Technical Product Management | Robotics and Manufacturing | AI-Enabled Products
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Robotics engineer and Columbia M.S. Technology Management candidate who evaluated 10 factories in China for U.S. customers including Toyota USA, deployed an LLM prototype, and filed a U.S. provisional patent application.

EXPERIENCE

SourceOne Ltd | *Product Manager (Part-time Remote to Full-time On-site)* Hong Kong | Feb to Aug 2026

- Evaluated **10 automated factories in China** on site for production capability, process controls, manufacturability, and supplier risk, inspecting casting, forging, machining, and laser-cutting against engineering drawings.
- Translated requirements from 3 U.S. customers (Mainspring Energy, **Toyota USA**, Rough Country) into prioritized technical issues, risks, and next steps for cross-functional engineering, quality, and sourcing teams.
- Presented each factory recommendation to customers and senior product leadership, backed by supplier evaluations, quality reports, and product-testing observations.

Independent Consulting | *Founder & Technology Consultant* Remote | Jun 2019 to Dec 2025

- Led **10+ software, web, and analytics products** for construction, legal, and retail businesses from client discovery and requirements through delivery and iteration.
- Grew a referral-based practice to 20+ clients on recurring retainers as their primary contact for requirements and delivery.

Stantec | *Automation Engineering Intern* Chandler, AZ | Jun to Aug 2024

- Caught discrepancies in 25% of 90 legacy electrical schematics across 12 mining-infrastructure projects and corrected the matching Revit models ahead of multidisciplinary design reviews.

Addictive Desert Designs | *New Product Development Intern* Mesa, AZ | Jun to Aug 2023

- Supported layout, fixturing, and controller setup for 2 OTC DAIHEN **robotic welding cells now in production**.
- Cut design cycles by **approximately 30%** with a fitment database covering 50+ truck models that reduced repeat research.

SELECTED PROJECTS

PerkPulse AI | *Creator & Product Lead, Columbia M.S. Project* | Live Prototype | GitHub May 2026

- Deployed an LLM prototype in Next.js, TypeScript, and Supabase PostgreSQL that turns dense credit card issuer guides into structured benefits data and monthly actions.
- Scored **20/20 valid parser outputs and 28/30 correct category detections** in controlled tests across 2 GPT-4o-mini workflows with confidence ratings, human review, and fallback logic.

FoldRide Pro | *Inventor & Product Lead, Foldable Electric Skateboard* 2025 to Present

- Tested the sub-$999 target price in concept feedback from 500 prospective riders, 70% of whom said they would buy.
- Filed a **U.S. provisional patent application**, took the design through 6 drawing revisions to production-intent STEP files, and planned the first prototype build, cost, and timeline with a factory partner.

Robotic Explorer for Hypothesized Surfaces | *NASA Psyche Student Collaboration | ASU Senior Capstone* Jan to Dec 2024

- Proposed the rover concept and **led the 4-person team** from system requirements through prototyping and testing, owning the full electrical subsystem from power distribution to motor control.
- Resolved a design risk raised in NASA Psyche mentor review by replacing hydraulic actuation with vacuum-compatible springs.
- Enabled 360-degree turns without reversing through independent wheel speed and direction control.

EDUCATION

Columbia University, M.S. Technology Management, School of Professional Studies Expected Dec 2026
GPA 3.9 | Coursework in Technology Product Management, AI/ML for Leaders, and Digital Transformation

Arizona State University, B.S. in Engineering, Robotics Engineering, focus in Electrical Engineering and Mechatronics Dec 2024

SKILLS

Product	Customer discovery | Requirements and PRDs | Prioritization | Stakeholder management | Product validation
Engineering	DFM | NPI | Supplier evaluation | Root cause analysis | BOM and costing | Mechatronics | Embedded systems
Tools	Python | SQL | C/C++ | TypeScript | OpenAI API | MATLAB | SolidWorks | Revit | AutoCAD | Jira | Git | Excel
Certifications	Google Foundations of Project Management (2026) | Google AI Essentials (2026)
Athletics	Three-time Ukrainian national junior swimming champion | Swim team captain, Lake Forest Academy
Languages	English, Russian, and Ukrainian (fluent)